



WORK SHEET - 33

DIFFERENTIAL EQUATION + TRIGONOMETRY

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.22 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is
(A) 3000 (B) 3500 (C) 4500 (D) 2500
- Q.2 The population $P(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dP(t)}{dt} = 0.5 P(t) - 450$. If $P(0) = 850$, then the time of which the population becomes zero, is
(A) $\frac{1}{2} \ln 18$ (B) $\ln 18$ (C) $2 \ln 18$ (D) $\ln 9$
- Q.3 A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is
(A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) $\frac{9}{7}$ (D) $\frac{7}{9}$
- Q.4 If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, the $y(\ln 2)$ is equal to
(A) 5 (B) 13 (C) -2 (D) 7
- Q.5 The solution of the differentiable equation $(1+y^2) + \left(x - e^{\tan^{-1}y}\right) \frac{dy}{dx} = 0$, is
(A) $(x-2) = K e^{-\tan^{-1}y}$ (B) $2x e^{\tan^{-1}y} = e^{2\tan^{-1}y} + K$
(C) $x e^{\tan^{-1}y} = \tan^{-1}y + K$ (D) $x e^{2\tan^{-1}y} = e^{\tan^{-1}y} + K$
(where k is constant of integration)
- Q.6 The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$ where a is an arbitrary constant is
(A) $2(x^2 - y^2)y' = xy$ (B) $2(x^2 + y^2)y' = xy$ (C) $(x^2 - y^2)y' = 2xy$ (D) $(x^2 + y^2)y' = 2xy$
- Q.7 The solution of the differential equation $ydx + (x + x^2y)dy = 0$, is
(A) $\frac{-1}{xy} = c$ (B) $\frac{-1}{xy} + \ln y = c$ (C) $\frac{1}{xy} + \ln y = c$ (D) $\ln y = cx$



- Q.8 The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants is
 (A) second order and second degree (B) first order and second degree
 (C) first order and first degree (D) second order and first degree
- Q.9 The differential equation of all circles passing through the origin and having their centres on the x-axis is
 (A) $x^2 = y^2 + xy \frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy \frac{dy}{dx}$ (C) $y^2 = x^2 + 2xy \frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy \frac{dy}{dx}$
- Q.10 The eccentricity of hyperbola satisfying the differential equation $2xy \frac{dy}{dx} = x^2 + y^2$, and passing through P(2, 1) is
 (A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) $3\sqrt{2}$ (D) 2
- Q.11 The general solution of the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$, is
 (A) $y^2 + 1 = x^2 + cy$ (B) $y^2 + 1 = x^2 + cx$ (C) $x^2 + 1 = y^2 + cy$ (D) $x^2 + 1 = y^2 + cx$
- Q.12 The general solution of $\frac{dy}{dx} = \sqrt{y-x}$ is equal to
 (A) $x = 2 \left(\sqrt{y-x} + \ln |\sqrt{y-x} - 1| \right) + c$ (B) $x = 2 \left(\sqrt{y-x} - \ln |\sqrt{y-x} - 1| \right) + c$
 (C) $y = 2 \left(2\sqrt{y-x} + \ln |\sqrt{y-x} - 1| \right) + c$ (D) $y = 2 \left(\sqrt{y-x} - \ln |\sqrt{y-x} - 1| \right) + c$
- Q.13 The interval $[a, b]$ such that the value of $\int_a^b (2 + x - x^2) dx$ is maximum, is
 (A) $[-1, 2]$ (B) $[-2, 1]$ (C) $[-2, -1]$ (D) $[1, 2]$
- Q.14 The complete set of values of x, satisfying the inequality $\cos 2x > |\sin x|$, where $x \in \left(-\frac{\pi}{2}, \pi \right)$ is
 (A) $\left(-\frac{\pi}{6}, \pi \right)$ (B) $\left(-\frac{\pi}{6}, \frac{\pi}{6} \right) \cup \left(\frac{5\pi}{6}, \pi \right)$
 (C) $\left(\frac{\pi}{6}, \frac{\pi}{2} \right) \cup \left(\frac{5\pi}{6}, \pi \right)$ (D) $\left(-\frac{\pi}{6}, \frac{5\pi}{6} \right)$
- Q.15 In triangle ABC, $a = 4$, $b = 3$ and the medians AA_1 and BB_1 are mutually perpendicular, then area (ΔABC) is equal to
 (A) $\sqrt{13}$ (B) $\sqrt{11}$ (C) $\sqrt{15}$ (D) $\sqrt{21}$

- Q.16 In ΔABC , if $a^2 + c^2 = 2002 b^2$, then $\frac{\cot A + \cot C}{\cot B}$ is equal to
 (A) $\frac{1}{2001}$ (B) $\frac{2}{2001}$ (C) $\frac{3}{2001}$ (D) $\frac{4}{2001}$
- Q.17 In ΔABC , if $\sin A + \sin B + \sin C = 1 + \sqrt{2}$ and $\cos A + \cos B + \cos C = \sqrt{2}$, then the triangle is
 (A) equilateral (B) isosceles (C) right angled (D) right angled isosceles
- Q.18 If in a triangle ABC, $\sin^4 A + \sin^4 B + \sin^4 C = \sin^2 B \sin^2 C + 2 \sin^2 C \sin^2 A + 2 \sin^2 A \sin^2 B$, then $\angle A$ is equal to
 (A) 30° (B) 120° (C) 60° (D) 45°
- Q.19 In triangle ABC, the incircle touches the sides BC, CA and AB respectively at D, E and F. If the radius of the incircle is 4 and BD, CE and AF are consecutive integers, then $s =$
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) 18 (B) 20 (C) 21 (D) 24
- Q.20 Let the angles A, B and C of ΔABC are in A.P. and $b : c = \sqrt{3} : \sqrt{2}$, then $\angle A =$
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) 45° (B) 75° (C) 105° (D) 110°
- Q.21 The set of all real numbers α such that $\alpha^2 + 2\alpha$, $2\alpha + 3$ and $\alpha^2 + 3\alpha + 8$ are the sides of a triangle is
 (A) $\alpha > 2$ (B) $\alpha > 3$ (C) $\alpha > 4$ (D) $\alpha > 5$
- Q.22 Tangent at any arbitrary point $p(x, y)$ on the curve $y = f(x)$ cuts the x -axis at A. 'B' is the foot of perpendicular drawn from P on x -axis. If the mid-point of AB always remains $(2, 0)$, then the curve is
 (A) $y^2 = c(x - 2)$ (B) $x^2 = c(y - 2)$ (C) $xy = c(x - 2)$ (D) $xy = c(y - 2)$

ANSWER KEY

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|------|---|------|---|------|---|------|---|------|---|------|---|------|---|
| Q.1 | B | Q.2 | C | Q.3 | A | Q.4 | D | Q.5 | B | Q.6 | C | Q.7 | B |
| Q.8 | D | Q.9 | C | Q.10 | B | Q.11 | B | Q.12 | A | Q.13 | A | Q.14 | B |
| Q.15 | B | Q.16 | B | Q.17 | D | Q.18 | A | Q.19 | C | Q.20 | B | Q.21 | D |
| Q.22 | A | | | | | | | | | | | | |

